National University of Computer and Emerging Sciences

Lahore Campus

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Data Communication Networks (EE2007)

Date: September 21st 2024

Course Instructor:

Dr. Saima Zafar

Sessional-I Exam[redacted]

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

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Attempt all the questions.

CLO # 1:

Q1: A. Answer the following Question. **[marks: 5]** 5

Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates $R1 = 500$ kbps, $R2 = 2$ Mbps, and $R3 = 1$ Mbps.

a. Assuming no other traffic in the network, what is the throughput for the file transfer?

b. Suppose the file is 4 million bytes. Roughly how long will it take to transfer the file to Host B?

Q1: B. Answer the following Question. **[marks: 5]** 2.5

- 1 Gb/s link — R
- each user:
  - 100 Mb/s when "active"
  - active 10% of the time

10

How many users can use this network under circuit-switching and packet-switching?
[In packet-switching, with 35 users, probability > 10 active users at the same time is less than .0004]

CLO # 2:

Q2: A. Answer the following Questions. **[marks: 5]** 5

A client and a Server support Cookie technology. The Server assigns a cookie # 1234 to the Client. The Client connects to the Server again after one week. Show how Cookie technology operates in this scenario with the help of a timing diagram. Clearly show ALL cookie components.

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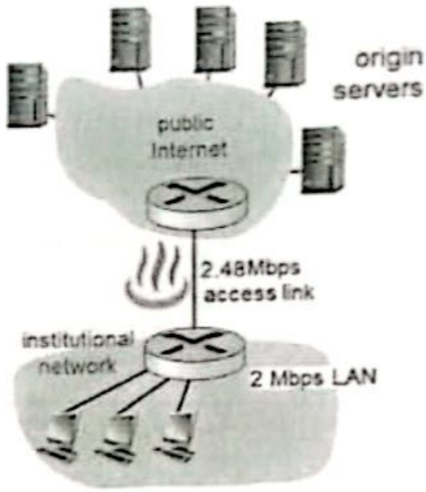
Q2: B. Answer the following Question. **[marks: 5]** 3

Consider the following scenario in the Figure.

- Access link rate: 2.48 Mbps
- RTT from institutional router to server: 2 sec
- web object size: 100K bits
- average request rate from browsers to origin servers: 12/sec
- avg data rate to browsers: 1.50 Mbps

Show working to find the estimated end-to-end delay. In light of your answer, tell what can be the solution(s) to minimize this delay? Justify your solution(s) by delay estimation after incorporating your solution(s).



CLO # 2:

Q3: A. Answer the following Questions. **[marks: 5]** 2

A client sends a DNS query to its local DNS server to get the IP address of nu.edu.pk. The local DNS server sends an iterative query to the TLD server which does not have the IP address of nu.edu.pk in its cache. It has the IP address: 128.118.4.3 of the Authoritative Server dns.edu.pk. The Authoritative DNS server has the IP address of nu.edu.pk: 125.30.3.111.

1. Draw a figure clearly showing the queries and replies for these iterative queries (Number the queries).
2. Write the Resource Record(s) in the TLD DNS server for this scenario.
3. Write the Resource Record(s) in the Authoritative DNS server for this scenario.

Q3: B. Answer the following Question. **[marks: 5]** 5

UDP and TCP use 1s complement for their checksums. Suppose you have the following three 8-bit bytes: 01010011, 01100110, 01110100. Compute the checksum. (Note that although UDP and TCP use 16-bit words in computing the checksum, for this problem you are being asked to consider 8-bit sums.) Show all work.